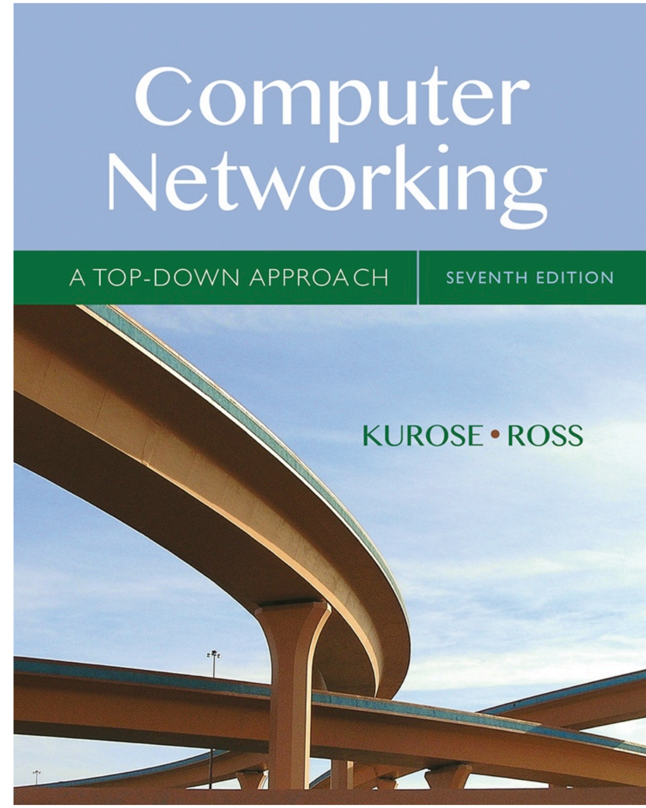
Chapter 1
Introduction



Computer Networking: A Top Down Approach

7th edition
Jim Kurose, Keith Ross
Pearson/Addison Wesley
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Chapter 1: roadmap

1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, network structure

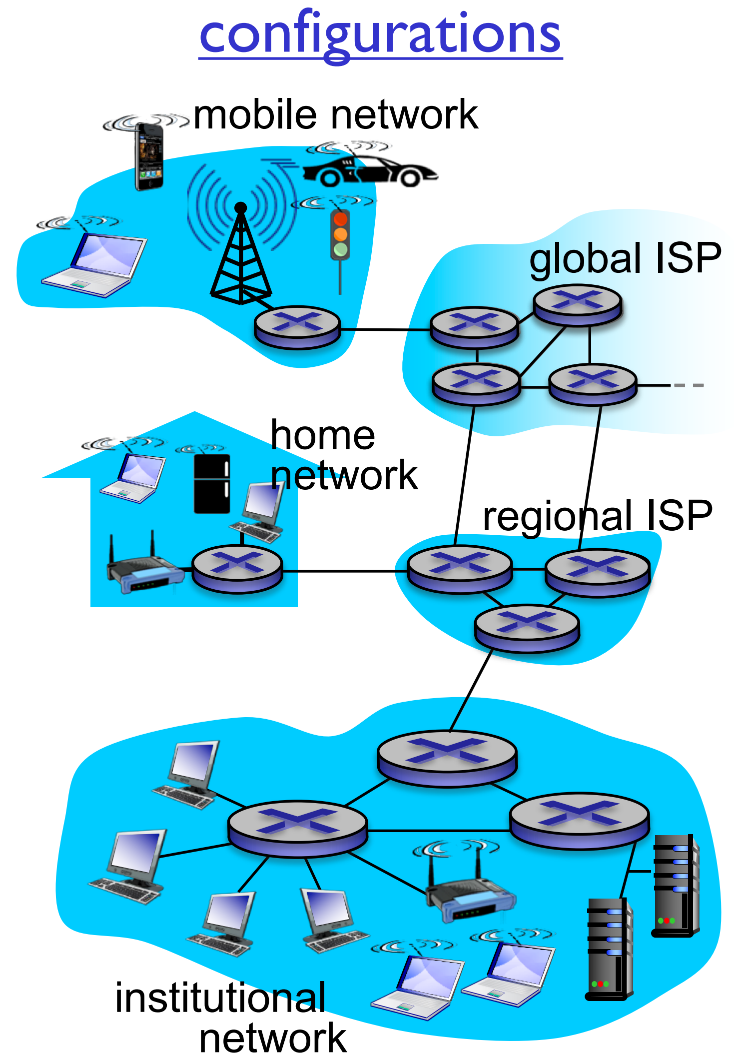
1.5 protocol layers, service models

What's the Internet: a "nuts and bolts" view

- billions of connected computing devices:
  - *hosts* = *end systems*
  - running *network apps*

- *communication links*
  - fiber, copper, radio, satellite
  - transmission rate: *bandwidth*

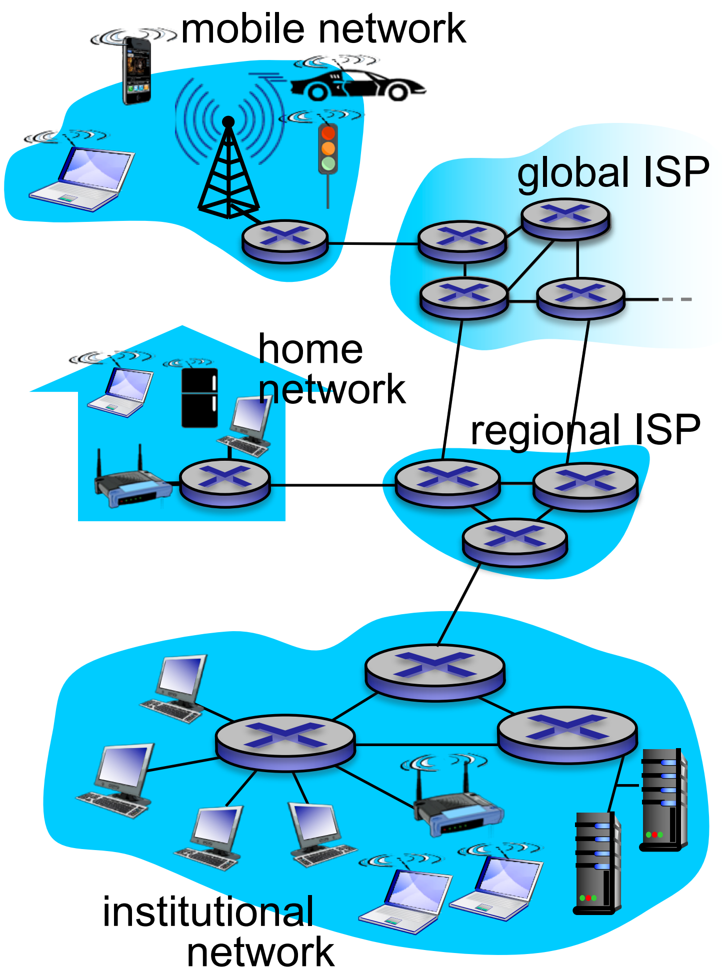
- *switches:* forward chunks of data (packets)
  - *routers* and *switches*



What's the Internet: a "nuts and bolts" view

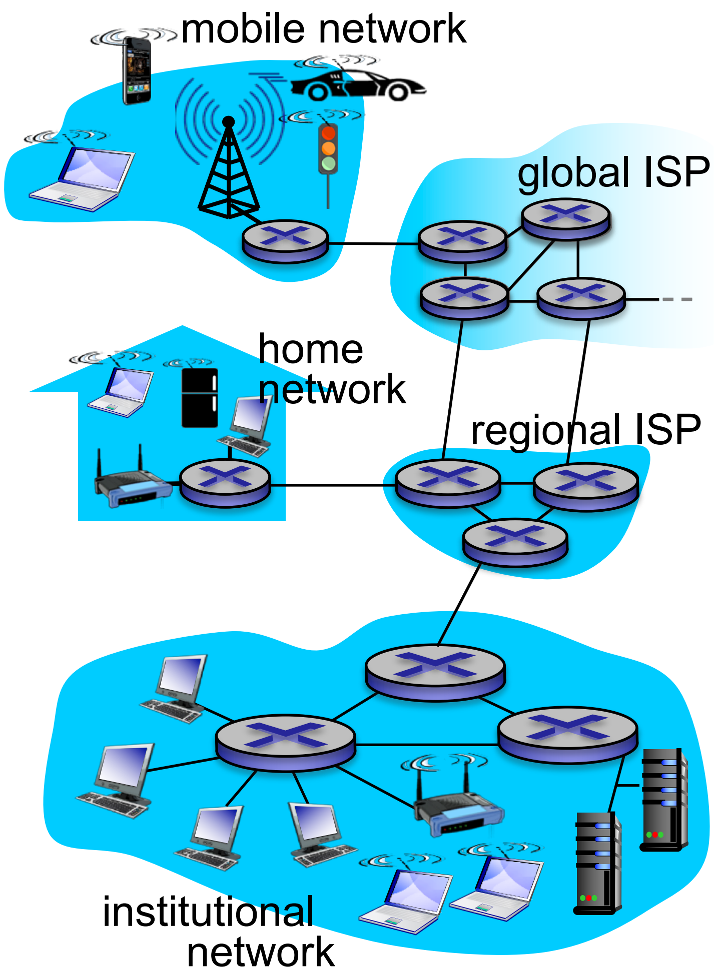
tying it all together

- *Internet* "network of networks"
  - interconnected networks transport data
- *protocols* control sending, receiving of information – data exchange between end devices
  - e.g., TCP, IP, HTTP, Skype, 802.11
- *Internet standards*
  - IETF: Internet Engineering Task Force
    - RFC: Request for comments
  - IEEE: Inst. of Electrical & Electronic Eng.
  - ITU: Intl Telecomm. Union (UN)



What's the Internet: a "service view"

- *an infrastructure that provides services to applications:*
  - Web, VoIP, email, games, e-commerce, social nets, …
- *it provides programming interface to apps*
  - software modules (socket) that allow sending and receiving application programs to "connect" to Internet
  - service options (options/choices) necessary for the functionality of applications



Postal Service Analogy

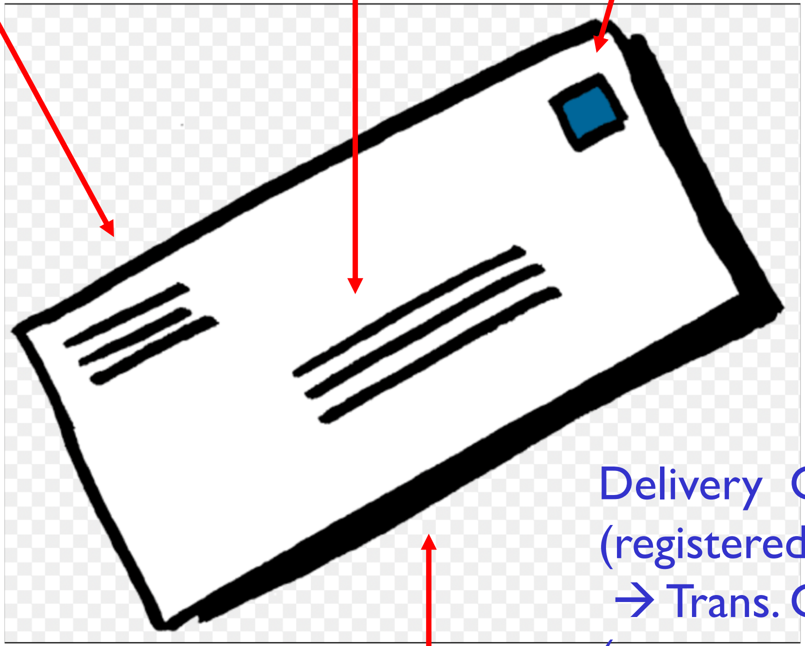
Postal Service Rules → Internet Rules

From address

To address

Stamp

Letter → Data



Delivery → Transmission

Delivery Options (registered, certified, priority) → Trans. Options (error control, priority)

Envelope → Packet

What's a protocol?

human protocols:

- asking a question
- in a classroom
- introductions
- a courtroom
- dinner conversation

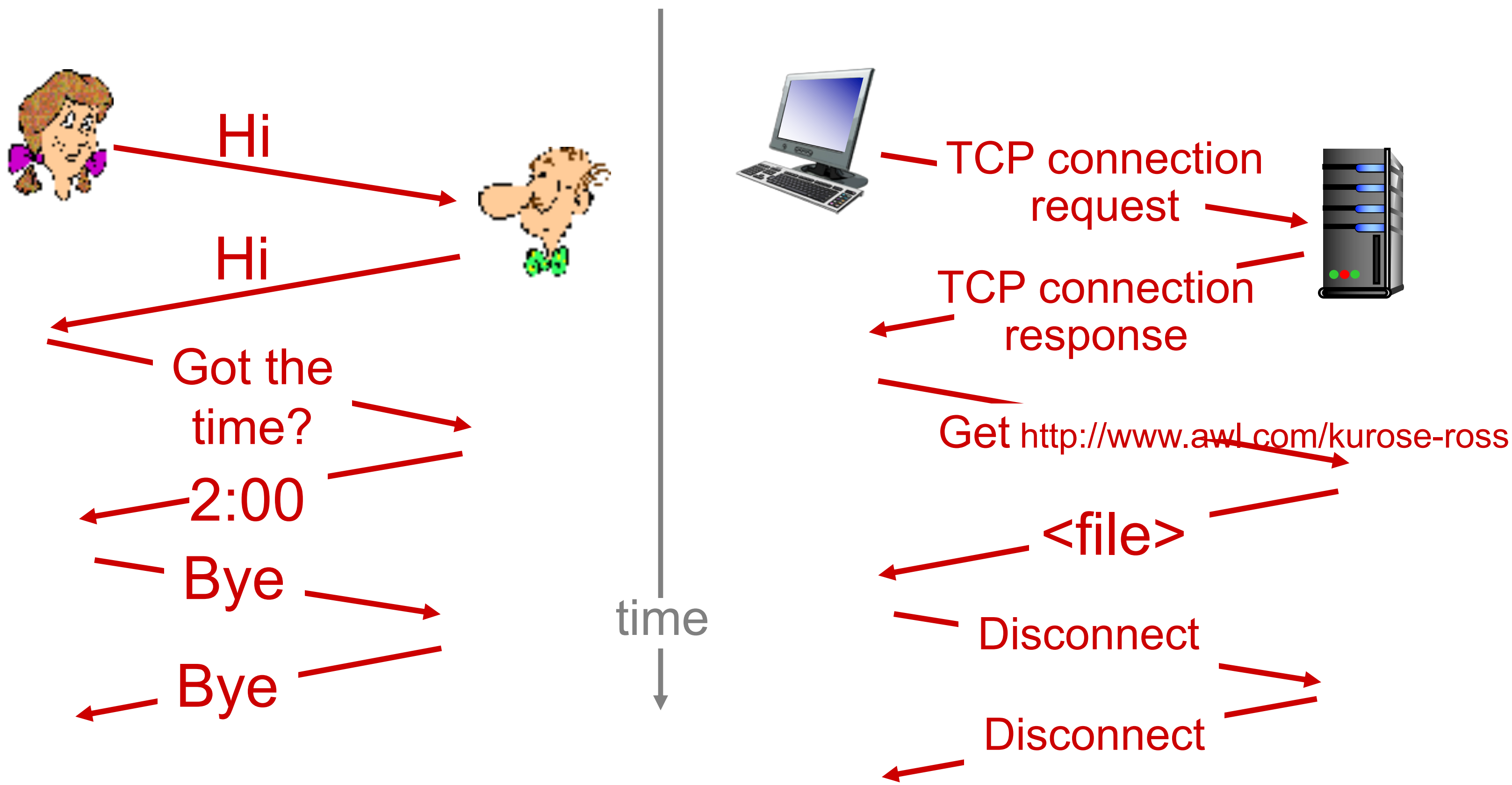
network protocols – similar but:

- devices rather than humans
- diversity of devices
- type of application:
  - queries
  - reports/files

protocols define format, order of messages sent and received and actions taken on message transmission, receipt

What’s a protocol?

a human protocol and a computer network protocol:



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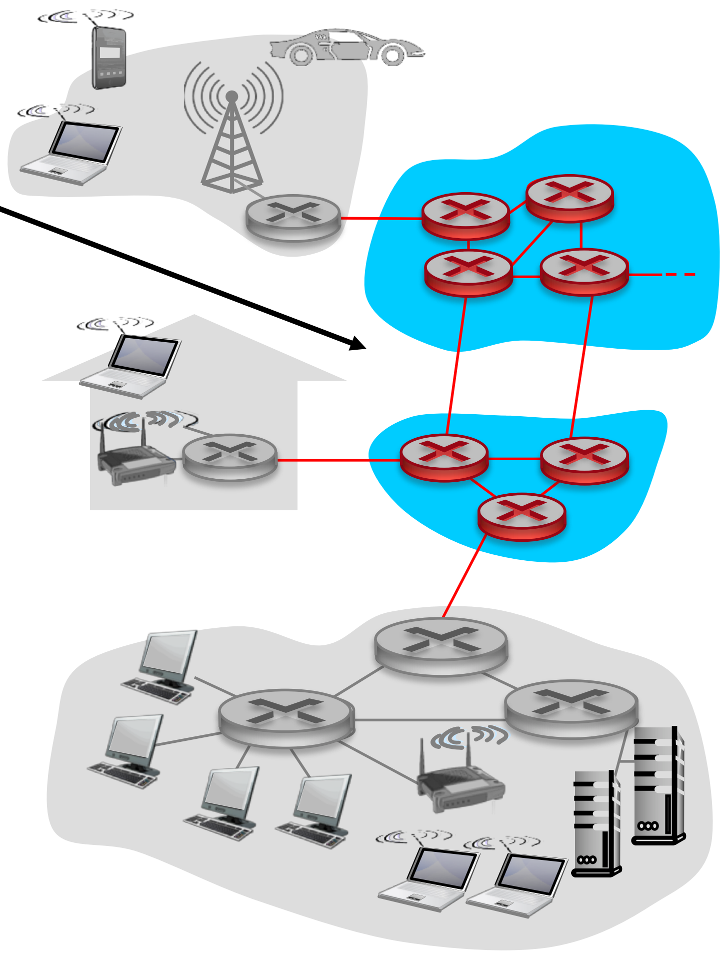
The network core

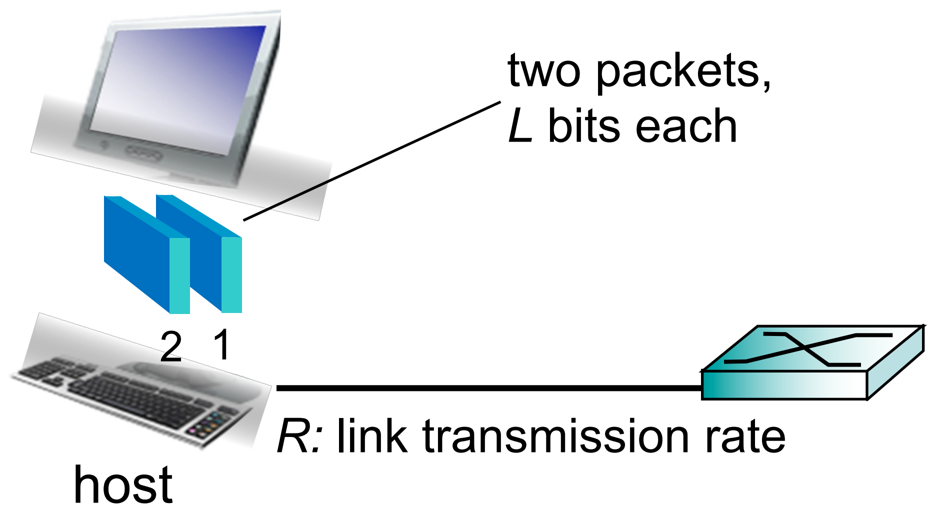
- mesh of interconnected switches referred to as *routers*
- packet-switching: hosts break application-layer data into chunks and put into *packets*
  - forward packets from one *router* to the next, across links on path from source to destination
  - each packet transmitted at full link capacity



End devices: generate data

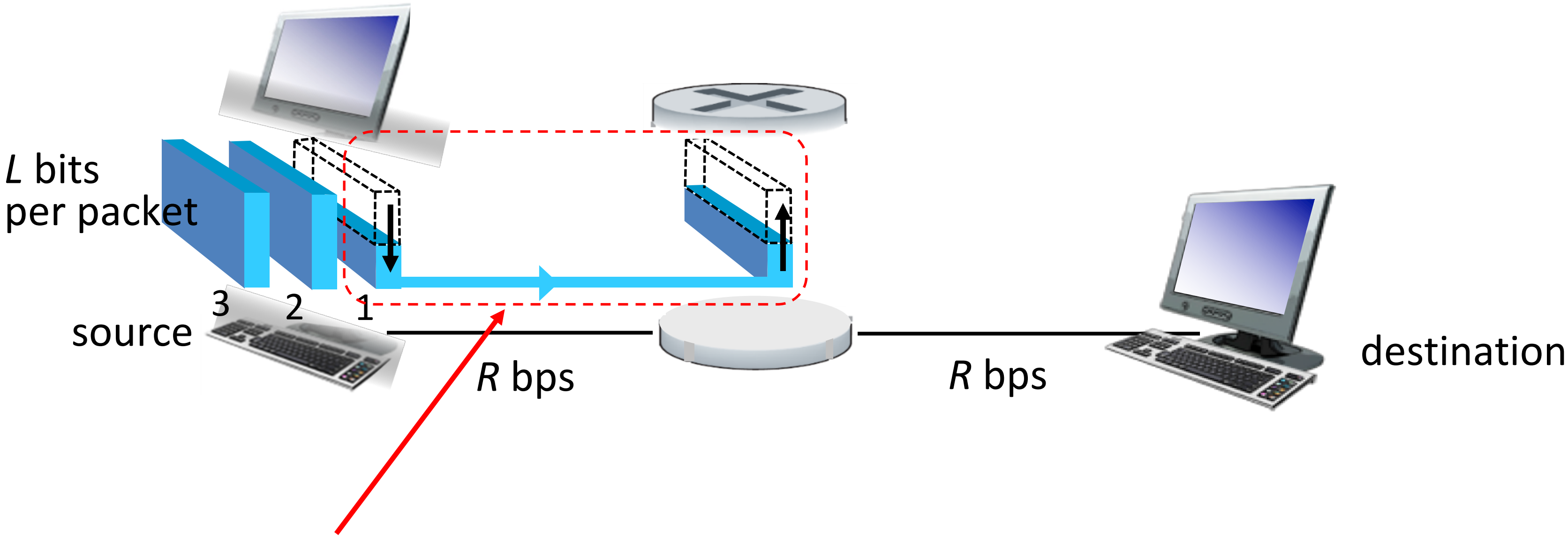
host sending function:

- takes application data
- breaks into smaller chunks, known as *packets*, of length *L* bits
- transmits packet into access network at *transmission rate R*
  - link transmission rate, aka *link capacity, aka link bandwidth*



$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet over link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Packet-switching: store-and-forward



- *store and forward:* entire packet must arrive at router before it can be transmitted on next link – *hop by hop*